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(1) Research Program/Project/Study Title: "Quantifying and Characterizing the Amount of Waste Generated in CAPSU Sigma Satellite College: Input to Waste Management Program of the Campus"	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

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Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:							
		Province:							
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(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
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(7) Sector/Commodity Waste Management – Waste	(8) Discipline/Field of Specialization Waste Management
(9A) Abstract Solid waste management posed significant challenges globally, including in the Philippines, where urban areas often struggled with inadequate infrastructure for waste handling and disposal. This study aimed to quantify and characterize waste generated at CAPSU Sigma Satellite College to develop sustainable waste management practices. The objectives included identifying waste sources and compositions, evaluating current waste management practices, and addressing associated challenges. Methodologies involved waste audits, which systematically sorted, weighed, and documented various types of waste, and surveys to capture data on waste sources and community attitudes towards waste management. The findings revealed that classrooms were the primary sources of waste, followed by the canteen, restrooms, and offices. The most common types of waste included plastic, paper, and food waste. Current waste management practices at the college included waste segregation at the source, recycling programs, composting, and landfill disposal. However, significant challenges were identified, such as a lack of awareness and participation among the college community, inadequate infrastructure, financial constraints, and limited space for waste disposal. The implications of this study emphasized the need for targeted waste reduction initiatives, particularly in classrooms, and the enhancement of recycling and composting programs. Addressing food and plastic waste was crucial for reducing the college's environmental footprint. Furthermore, this research was conducted in alignment with Sustainable Development Goal (SDG) 11, which focuses on creating sustainable cities and communities. By improving waste management practices, the college could promote a healthier campus environment and contribute to broader sustainable development goals.	
(9B) Keywords: <i>Sustainable Development Goal, Waste Management Practices, Waste Sources, Waste Composition, Recycling Practices, Sustainability Challenges</i>	
(10) Introduction Solid waste management is a pressing issue globally, including in the Philippines, where issues of waste generation, handling, and disposal pose significant challenges, particularly in densely populated urban areas with inadequate infrastructure and poor disposal practices. In this context, our study aimed to address these challenges by quantifying and characterizing the waste generated at CAPSU Sigma Satellite College, a tertiary institution in the Philippines. Through this research, the researchers sought to gain a comprehensive understanding of the waste generation patterns within the college and develop recommendations for more efficient and sustainable waste management practices. Korstad and Yee (2018) highlighted the challenges facing municipal solid waste management in the Philippines, emphasizing issues such as inadequate infrastructure and governance, as well as the significant role of informal waste pickers in the country's waste management system. Their recommendations, including the development of integrated waste management plans and stricter regulations, serve as valuable insights for this study. Moreover, the research aligns with the Philippines' focus on research and development (R&D) as a means to address social and economic challenges, including environmental sustainability. The feasibility study conducted by the Institute for Global Environmental Strategies (2015) on waste-to-energy potential in the Philippines underscores the importance of exploring innovative solutions to waste management issues. Additionally, Bautista and Abalos (2016) shed light on the current state of solid waste management practices in Philippine higher education institutions, highlighting the need for improved waste segregation and disposal methods. The study aimed to address the lack of comprehensive data on waste generation in CAPSU Sigma Satellite College by answering key questions regarding the awareness, sources, composition, and total amount of waste generated, as well as the current waste management practices and associated challenges. By providing this information, we aim to contribute to the development of evidence-based policies and programs for sustainable waste management practices in educational institutions.	

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The significance of the study lies in its potential environmental impact, as it will inform strategies for reducing the college's environmental footprint through improved waste management. Additionally, by addressing health and sanitation issues associated with waste disposal, the research will contribute to creating a safer and healthier campus environment. Furthermore, by identifying opportunities for resource management and policy development, the study will support efficient resource utilization and the development of effective waste management practices.

Through stakeholder engagement and the utilization of technologies such as waste management software and bioreactors, we aim to propose recommendations tailored to the specific needs of CAPSU Sigma Satellite College. By disseminating the findings to policymakers, waste management practitioners, educators, students, and the general public, we seek to foster awareness and facilitate the implementation of sustainable waste management practices.

In conclusion, this study on waste quantification and characterization at CAPSU Sigma Satellite College was not only justified in terms of resource expenditure but also directly addressed Sustainable Development Goal 11 (Sustainable Cities and Communities). This research is crucial for tackling pressing environmental challenges and promoting the well-being of the college community and beyond.

(11) Materials and Methods or Methodology

The study "Quantifying and Characterizing the Amount of Waste Generated in CAPSU Sigma Campus: Input to Waste Management Program of the Campus" aimed to explore the dynamics of waste within the campus environment by applying fundamental concepts such as waste management, waste generation, and recycling. The study focused on key variables, including waste generation rate, waste composition, attitudes towards waste reduction, and recycling rate, to understand waste production and management practices on campus.

The methodology involved a comprehensive data collection approach. Waste audits were conducted to accurately quantify and categorize different types of waste generated across the campus, involving systematic sorting, weighing, and documentation. Additionally, surveys were administered to students, faculty, and staff to gather detailed data on waste sources and current waste management practices, providing insights into community attitudes and behaviors towards waste reduction and recycling.

Data analysis employed descriptive statistics to summarize the collected information, using means, percentages, and frequencies to provide a clear picture of waste generation patterns and management practices. This analysis helped identified trends and relationships within the data, laying the groundwork for evaluating current waste management strategies and pinpointing areas for improvement.

The research design was descriptive, relying on quantitative data from waste audits and surveys to offer a thorough examination of waste generation patterns and management practices. This design supported a detailed understanding of the effectiveness of current waste management practices and the potential for enhancements.

Data gathering for the study spans from August 8, 2022, to January 8, 2023, ensuring a comprehensive assessment of waste management practices over an extended period. The implementation phase includes regular waste audits and surveys to continuously monitor waste generation and management practices. Based on the identified trends and patterns, recommendations for improvement were developed, and the effectiveness of these practices was ensured through ongoing evaluation and stakeholder feedback mechanisms. This approach ensured that the waste management program remained responsive and effective in addressing the campus's needs.

(12) Results and Discussions

Demographic Profile of Respondents
Table 1. Demographic Profile of Respondents

Role/ Position	F	%	Department/ Program	F	%	Age	F	%	Gender	F	%
Student	30	88	Hospitality Management	22	65	18-20	10	31	Female	24	72
Faculty	34	10	Tourism Management	55	16	21-23	17	49	Male	94	27
Staff	8	2	Teacher Education Administrations Office	58	17	24-26	27	8	Non-Binary		
				8	2	27-29	2	1	Prefer not to say	3	1

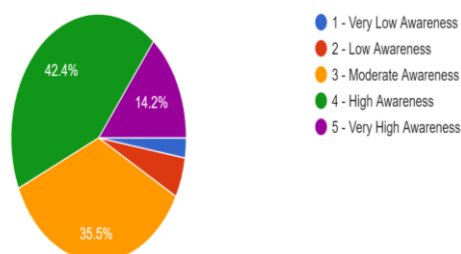
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				30 and Above	36	11		
34	10		34	10	34	10	34	10
4	0		4	0	4	0	4	0

The data provided offers insights into the demographics and distribution within a particular institution, likely an educational or training facility specializing in hospitality and tourism management. The majority of individuals, accounted for 88%, were students, primarily aged between 18 to 20 years, with a notable dominance of females at 72%. Faculty members represent 10% of the population, with males comprising 49% of this group. Staff members make up the smallest proportion at 2%, with a diverse distribution across departments, including Teacher Education and Administrative Office. The data also reflected a range of age groups, with a significant concentration in the 21 to 23 age bracket. Overall, the data provided a snapshot of the institution's demographics, illustrating a student-centric environment with notable gender imbalances across different roles and age groups.

Level of Awareness on Waste Management Practices within CAPSU Sigma Satellite College

Figure 1. Level of Awareness on Waste Management Practices within CAPSU Sigma Satellite College



The pie chart illustrated the level of awareness on waste management practices among respondents at CAPSU Sigma Satellite College. The data revealed a spectrum of awareness levels, with the majority of respondents displaying a higher degree of awareness. Specifically, 42.4% of the participants indicated a high awareness (represented in green), while 14.2% demonstrated very high awareness (depicted in purple). This suggested that over half of the respondents possessed a significant understanding of waste management practices.

Moderate awareness (represented in orange) was reported by 35.5% of the participants, indicating that a substantial portion of the community has a reasonable but not profound understanding of these practices. In contrast, low awareness (represented in red) and very low awareness (represented in blue) were reported by 5% and 2.9% of respondents, respectively. This highlights that a small fraction of the population is either unaware or only slightly aware of waste management practices.

The implications of these findings were noteworthy. The high level of awareness observed among most respondents suggested effective communication and educational efforts regarding waste management within the college. However, the presence of individuals with low and very low awareness points to the need for targeted educational interventions to ensure comprehensive understanding across the entire community. Addressing these gaps can further enhance the overall efficacy of waste management initiatives at CAPSU Sigma Satellite College, fostering a more environmentally conscious campus culture.

Sources of Waste in CAPSU Sigma Satellite College

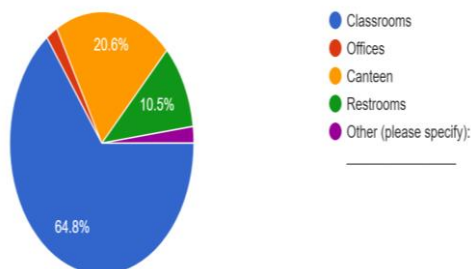


Figure 2. Sources of waste in CAPSU Sigma Satellite College

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The pie chart depicts the primary sources of waste within CAPSU Sigma Satellite College. The data highlighted that the majority of waste were generated from classrooms, accounted for 64.8% (represented in blue). This significant proportion suggests that a large volume of the waste produced within the college is likely paper, plastic, and other materials associated with classroom activities and student usage.

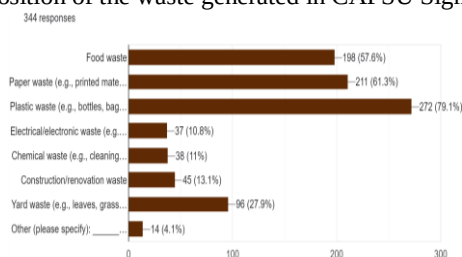
Canteen was the second-largest source of waste, contributing 20.6% (represented in orange). This indicated that food waste, packaging, and other related materials were substantial contributors to the overall waste output. Restrooms account for 10.5% of the waste (represented in green), which typically includes paper towels, hygiene products, and related items. Offices contributed 2.9% of the waste (represented in red), reflecting a smaller yet notable amount of waste likely comprising paper, office supplies, and packaging materials. The "Other" category accounts for a minimal 1.2% (represented in purple), indicating a small fraction of waste from miscellaneous sources not specified in the main categories.

The implications of these findings were significant for developing targeted waste management strategies within the college. Given that classrooms are the predominant source of waste, initiatives such as promoting digital submissions, reducing paper usage, and increasing recycling bins in these areas could effectively reduce waste volume. The canteen's substantial contribution to waste generation suggested the need for strategies like composting food waste, reducing single-use plastics, and encouraging the use of reusable containers.

Efforts to minimize restroom waste could focus on installing efficient hand dryers to replace paper towels and promoting the use of eco-friendly hygiene products. Office waste reduction might be achieved through digitization and efficient recycling programs. Addressing the minimal "Other" category waste involved identifying and managing these miscellaneous sources effectively.

Overall, a comprehensive and nuanced approach that addresses the specific waste sources identified can significantly enhance the sustainability and waste management practices at CAPSU Sigma Satellite College, fostering a more environmentally responsible campus.

Composition of the waste generated in CAPSU Sigma Satellite College
Figure 3. Composition of the waste generated in CAPSU Sigma Satellite College



The bar chart presented the composition of waste generated at CAPSU Sigma Satellite College based on 344 responses. The data revealed that the most significant portion of waste is plastic waste, with 272 respondents (79.1%) identifying it as a major component. This was followed by paper waste, noted by 211 respondents (61.3%), and food waste, mentioned by 198 respondents (57.6%). These three categories—plastic, paper, and food waste—constitute the bulk of waste generated within the college.

Yard waste, such as leaves and grass, accounted for 27.9% (96 respondents), indicating a notable but smaller proportion. Construction and renovation waste is identified by 13.1% of respondents (45 responses), suggesting periodic waste from maintenance and building activities. Chemical waste, such as cleaning agents, was recognized by 11% of respondents (38 responses), while electrical/electronic waste is slightly lower at 10.8% (37 responses). The "Other" category, which included miscellaneous waste not specified in the primary categories, is the smallest, with 4.1% (14 respondents).

The implications of these findings highlighted critical areas for targeted waste management efforts. The prevalence of plastic waste suggested a need for initiatives to reduced plastic use, promoted recycling, and encouraged alternatives like reusable containers and packaging. The significant amount of paper waste indicate potential benefits from digitalization, double-sided printing, and enhanced paper recycling programs.

Food waste being a major component suggested implementing composting programs and strategies to reduced food waste in the canteen. Yard waste management involve composting and utilizing green waste for campus landscaping.

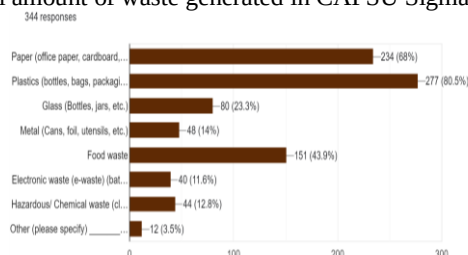
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The presence of construction and renovation waste, although less frequent, necessitates effective recycling and disposal practices during building projects.

The smaller proportions of chemical and electronic waste indicated areas where specific disposal protocols could be enforced to prevent environmental harm. The minimal "Other" category points to a relatively low occurrence of unspecified waste, but attention to detail in waste categorization and management was still essential.

Overall, a comprehensive waste management strategy at CAPSU Sigma Satellite College could prioritize reducing plastic and paper waste, enhanced recycling and composting programs, and ensuring proper disposal of less common waste types. These measures contributed to a more sustainable and environmentally responsible campus.

Total amount of waste generated in CAPSU Sigma Satellite College
Figure 4. Total amount of waste generated in CAPSU Sigma Satellite College



The total waste generated at CAPSU Sigma Satellite College was approximately 300 pounds, with a detailed breakdown as follows: 234 pounds (68%) of paper, 277 pounds (80.5%) of plastics, 80 pounds (23.3%) of glass, 48 pounds (14%) of metal, 151 pounds (43.9%) of food waste, 40 pounds (11.6%) of electronic waste (e-waste), 44 pounds (12.8%) of hazardous/chemical waste, and 12 pounds (3.5%) of other waste. This data highlighted several key insights into the college's waste generation patterns. The high volumes of plastics and paper, comprising 277 pounds (80.5%) and 234 pounds (68%) respectively, indicated a substantial reliance on disposable materials, with significant environmental impacts due to the non-biodegradable nature of plastics. Food waste is also considerable at 151 pounds (43.9%), suggesting inefficiencies in food management. Hazardous and electronic waste, totaling 44 pounds (12.8%) and 40 pounds (11.6%) respectively, require careful handling to prevent environmental contamination. Glass and metal wastes, though lower in volume at 80 pounds (23.3%) and 48 pounds (14%), present opportunities for recycling initiatives. To manage and reduce waste, the college could enhance recycling programs for paper and plastics, implement composting and better food management to reduce food waste, and established designated disposal points for hazardous and e-waste. Additionally, encouraging the use of reusable items and promoting digital solutions to minimize paper usage can further decrease waste generation. Reviewing the scope and accuracy of the data and assessing the effectiveness of existing waste management programs were crucial for developing comprehensive and sustainable waste reduction strategies. Overall, CAPSU Sigma Satellite College can significantly reduce its environmental impact by focusing on these areas.

Total amount of waste generated in CAPSU Sigma Satellite College
Table 2. Total amount of waste generated in CAPSU Sigma Satellite College

Waste Type	Daily (kg)	Weekly (kg)	Monthly (kg)	Annual (kg)
Paper	57.6 kg	440.41 kg	2891.1 kg	3,3791.22 kg
Plastics	80.71 kg	620.58 kg	3,949.9 kg	4,7398.8 kg
Glass	12.33 kg	96.05 kg	395.57 kg	4,746.84 kg
Metal	15.9 kg	125.16 kg	515.55 kg	6,186.6 kg
Food	87.94 kg	691.24 kg	2,845.22 kg	34,142.64 kg
Electronic	3.9 kg	30.71 kg	126.45 kg	1,517.40 kg
Hazardous	1.20 kg	9.46 kg	39 kg	468.02 kg

The table presented an overview of the total waste generated in CAPSU Sigma Satellite College across various waste types, delineated into daily, weekly, monthly, and annual quantities. From the data, two major findings emerged: Firstly, the significant dominance of food waste was evident, with a daily output of 87.94 kg, resulting in an annual total of 34,142.64 kg. This substantial amount underscores the imperative for targeted strategies aimed at reducing and diverting food waste within the college environment. Secondly, the noteworthy accumulation of plastic waste was

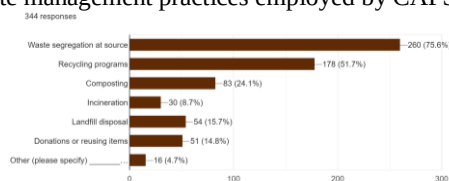
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apparent, with daily quantities reaching 80.71 kg, culminating in an annual total of 4,7398.8 kg. This highlighted the urgent need for initiatives aimed at curbing plastic consumption, bolstering recycling efforts, and exploring viable alternatives to single-use plastics.

These findings underscored the critical necessity for CAPSU Sigma Satellite College to implement comprehensive waste management strategies encompassing waste reduction, robust recycling programs, and adoption of sustainable practices. Such measures were essential in mitigating the environmental impact of waste generation and fostering a more environmentally responsible campus community.

Current waste management practices employed by CAPSU Sigma Satellite College

Figure 5. Current waste management practices employed by CAPSU Sigma Satellite College

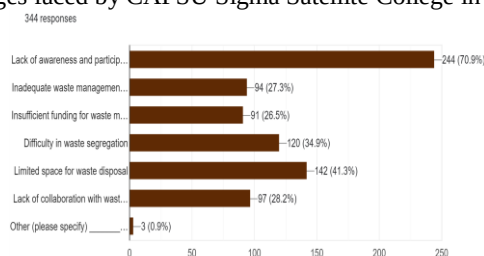


The current waste management practices at CAPSU Sigma Satellite College were as follows: waste segregation at source (75.6%), recycling programs (51.7%), composting (24.1%), landfill disposal (15.7%), donations or reusing items (14.8%), incineration (8.7%), and other methods (4.7%). The data indicated a strong emphasis on waste segregation at the source, which was practiced by 75.6% of the community. This was crucial for facilitating efficient recycling and composting, allowing the college to manage waste more effectively. Recycling programs, involving 51.7% of the efforts, demonstrated a commitment to diverting waste from landfills and reducing environmental impact. Composting, utilized by 24.1%, helped managed organic waste, reducing landfill volume and creating compost for gardening and landscaping. Despite these efforts, 15.7% of waste still ends up in landfills, indicating room for improvement in waste diversion strategies. Donations or reusing items (14.8%) extend the life of materials and reduce waste generation, while incineration, used by 8.7%, likely reflected concerns about air pollution and environmental impact.

The implications for waste management were clear: while the college has a strong foundation in waste segregation and recycling, there were opportunities to enhance these practices further. Increasing recycling and composting rates can be achieved by expanding educational campaigns and providing more resources. Reducing landfill waste could involved stricter waste segregation policies, increased recycling and composting infrastructure, and encouraging material reduction and reuse. Ensuring comprehensive waste segregation that included all types of waste can improve the effectiveness of recycling and composting programs. Regular assessments of waste management practices can help identify areas for improvement and ensure that the college meets its waste reduction goals. Additionally, the data did not specified the types of waste being recycled or composted, which would be useful for tailoring strategies. There may also be other waste management practices not captured in the data that could contribute to the college's efforts. Overall, CAPSU Sigma Satellite College showed a commitment to effective waste management, but there is potential for further improvement by focusing on increasing recycling and composting rates, reducing landfill disposal, and enhancing waste segregation. By addressing these areas, the college can continue to reduce its environmental impact and promote sustainability.

Challenges faced by CAPSU Sigma Satellite College in managing its waste

Figure 6. Challenges faced by CAPSU Sigma Satellite College in managing its waste



CAPSU Sigma Satellite College encountered several challenges in effectively managing its waste, as identified by respondents. Foremost among these challenges was a lack of awareness and participation, cited by 70.9% of respondents, underscoring the need for educational initiatives to instill responsible waste management practices within the college community. Additionally, inadequate waste management infrastructure, reported by 27.3% of respondents,

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poses hindrances to waste collection, segregation, and recycling efforts, necessitating upgrades or expansions. Financial constraints further exacerbate the situation, with 26.5% of respondents citing insufficient funding for waste management, prompting the exploration of alternative financing avenues. Space limitations for waste disposal, mentioned by 41.3% of respondents, present logistical challenges, particularly in the absence of robust recycling or composting options. Moreover, the lack of collaboration with waste management companies, acknowledged by 28.2% of respondents, suggests untapped opportunities for enhancing waste handling efficiency through strategic partnerships. Addressing these challenges demands a comprehensive approach involving educational campaigns, infrastructure enhancements, funding diversification, space optimization, and collaboration with external stakeholders. Additionally, tackling specific issues such as waste segregation difficulties and hazardous waste management was vital for achieving comprehensive waste management at CAPSU Sigma Satellite College.

(13) Conclusions and Recommendations

Conclusions

Based on the findings of the study, the following conclusions have been drawn:

1. The data highlighted a significant lack of awareness regarding waste management within the college community, but the presence of moderately and highly aware groups indicates potential for improvement through targeted education and awareness campaigns.
2. Analysis showed that classrooms were the main contributors to waste generation, primarily from discarded paper, food, and plastics, while contributions from offices and other sources highlighted the diverse nature of campus waste.
3. Food waste was the predominant waste stream, followed closely by paper, with minimal plastic waste, alongside significant streams such as electrical/electronic, chemical, and yard waste, reflecting the broad spectrum of waste generated within the college.
4. While the data did not provide a direct measure of total waste, it outlines waste composition, emphasizing significant quantities of food and plastic waste, necessitating urgent action due to their environmental impact.
5. Waste segregation at the source was the primary practice within the college community, although the effectiveness and scope of recycling programs remain unclear, with composting, incineration, landfill disposal, and donations/reuse initiatives also present in the college's waste management practices.
6. The foremost challenge identified was the lack of awareness and participation among the college community, compounded by inadequate infrastructure, funding constraints, and space limitations, with limited collaboration with waste management companies indicating missed opportunities for improving waste management processes.

Recommendations

Based on the conclusions of the study, the following recommendations have been drawn:

1. Implement targeted education and awareness campaigns to address the lack of awareness regarding waste management within the college community.
2. Initiate waste reduction initiatives specifically targeting classrooms to mitigate waste generation; to develop strategies to address the predominant food waste stream.
3. Take immediate action to address the significant quantities of food and plastic waste highlighted in the data.
4. Enhance the effectiveness and scope of recycling programs by providing clear guidelines on waste segregation, improving recycling infrastructure, and increasing awareness about the importance of recycling among the college community.
5. Address identified challenges by prioritizing initiatives to increase awareness and participation in waste management practices.

(14) Output

Waste Management Program for CAPSU Sigma Satellite College

Objectives

1. Reduce overall waste generation.
2. Increase waste segregation, recycling, and composting.
3. Enhance awareness and participation in waste management practices.
4. Address infrastructure and logistical challenges.

Program Components

1. Education and Awareness Campaigns

- **Target Audience:** Students, Faculty, Staff

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- **Content:** Importance of waste management, how to segregate waste, benefits of recycling and composting.
- **Methods:** Workshops, seminars, posters, social media campaigns, integrating into the curriculum.

2. Waste Reduction Initiatives

- **Classroom Strategies:**
 - Promote digital submissions to reduce paper waste.
 - Install more recycling bins for paper and plastics.
- **Canteen Strategies:**
 - Implement food waste reduction programs.
 - Encourage use of reusable containers and cutlery.

3. Enhanced Recycling and Composting Programs

- **Recycling:**
 - Clear guidelines on waste segregation.
 - Improved recycling infrastructure with more bins and regular collection.
- **Composting:**
 - Composting bins for food and yard waste.
 - Use compost for campus landscaping.

4. Infrastructure and Logistical Improvements

- **Upgraded Facilities:**
 - More waste segregation points.
 - Dedicated areas for hazardous and electronic waste.
- **Collaboration:**
 - Partner with local waste management companies for better waste handling and recycling processes.

Implementation Plan

Phase 1: Awareness and Education

- **Timeline:** 1-3 months
- **Activities:**
 - Launch awareness campaigns.
 - Conduct workshops and seminars.
 - Distribute educational materials.

Phase 2: Infrastructure Development

- **Timeline:** 3-6 months
- **Activities:**
 - Install new waste segregation bins.
 - Set up composting stations.
 - Improve recycling facilities.

Phase 3: Program Rollout

- **Timeline:** 6-12 months
- **Activities:**
 - Implement waste reduction initiatives in classrooms and canteens.
 - Begin composting and enhanced recycling.
 - Monitor and evaluate the effectiveness of the programs.

Phase 4: Continuous Improvement

- **Timeline:** Ongoing
- **Activities:**
 - Regularly assess program effectiveness.
 - Update and improve educational materials.
 - Enhance infrastructure as needed.

Monitoring and Evaluation

- **Metrics:**
 - Reduction in overall waste generation.
 - Increase in segregated and recycled waste.
 - Participation rates in awareness programs.
- **Methods:**
 - Monthly waste audits.
 - Surveys to gauge awareness and participation.
 - Feedback sessions with stakeholders.

(15) Technology Transfer Activity Conducted

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(16) Economic Analysis

(17) Acknowledgments

The researchers express sincere gratitude to God for His guidance and to their family for THEIR unwavering support. They also extend heartfelt thanks to friends and the university community for enhancing their journey. Additionally, appreciation is conveyed to the faculty and staff for their invaluable contributions.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

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